Analytical Chemistry 1 (125101)

Guangdong Technion
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Framework of this course

❖ Introduction (L1)

❖ Section I Chemical Measurements

1.1 Basic knowledge (L2)

1.2 Error and estimation (L3&L4)

1.3 Linear regression (L5)

❖ Section II Chemical Equilibrium

2.1 pH and buffer solutions (L6)

2.2 Titrations (L7&L8)

2.3 Potentiometry in titrations (L9)

2.4 Gran plot in titrations (L10)

2.5 Electroanalysis (L11)

2.6 Gravimetric analysis (L12)

❖ Summary (L13)

L13 Summary of

Analytical Chemistry 1

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What is Analytical Chemistry (AC)?

Develops & applies

- **Methods**
- **Instruments**
- **Strategies**

$\Longrightarrow$

1. **Titration**
2. **Electroanalyis**
3. **Gravimetry**

Information on **nature** and **composition** in space or time

$\Downarrow$ $\Downarrow$ $\Downarrow$ $\Downarrow$

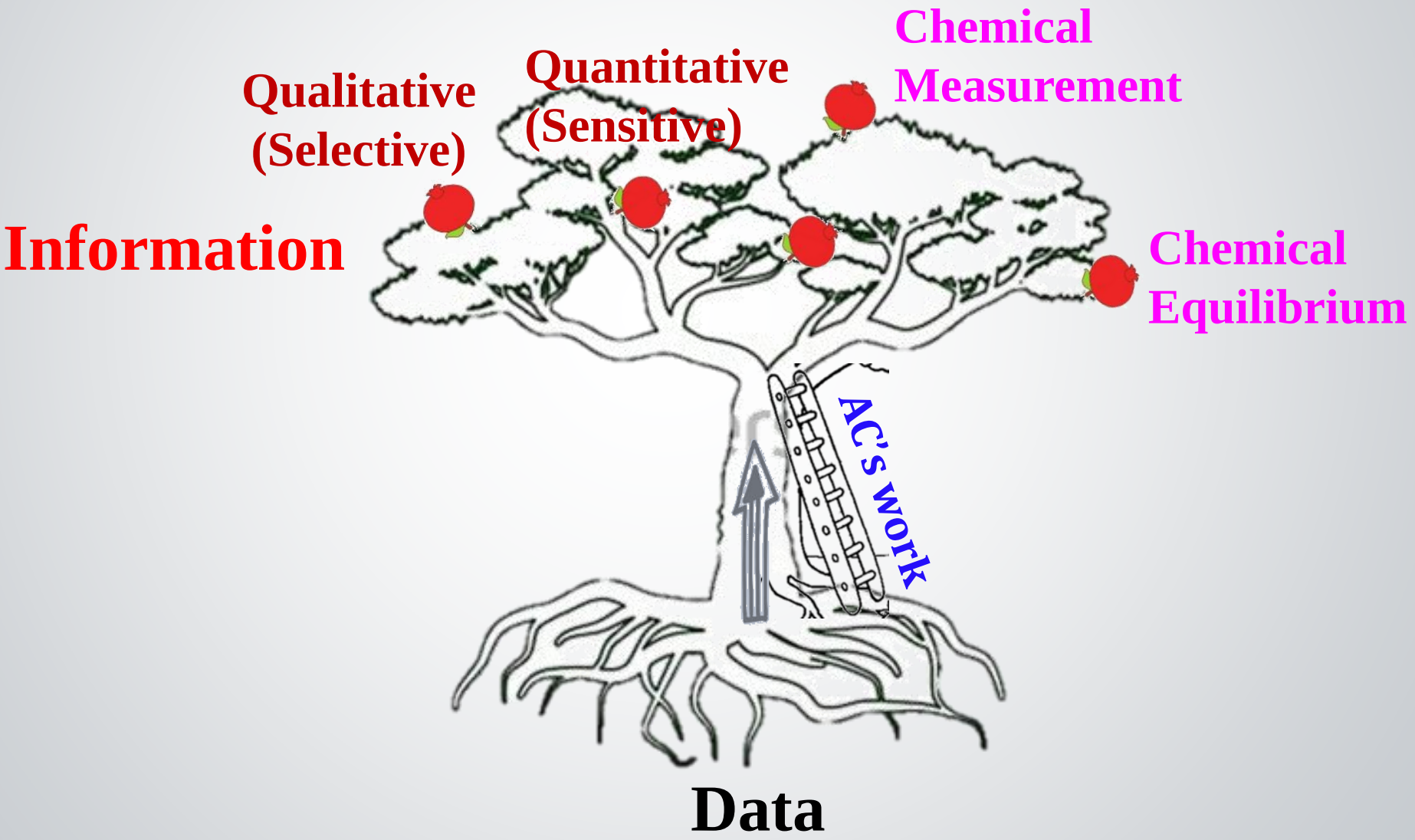
Data $\subset$ **Qualitative** + **Quantitative**

What is it ? **How much is it ?**

Chemical Measurements

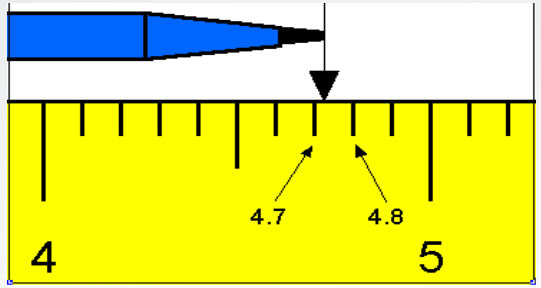
Chemical Equilibrium

What is Analytical Chemistry (AC)?



Chemical measurement

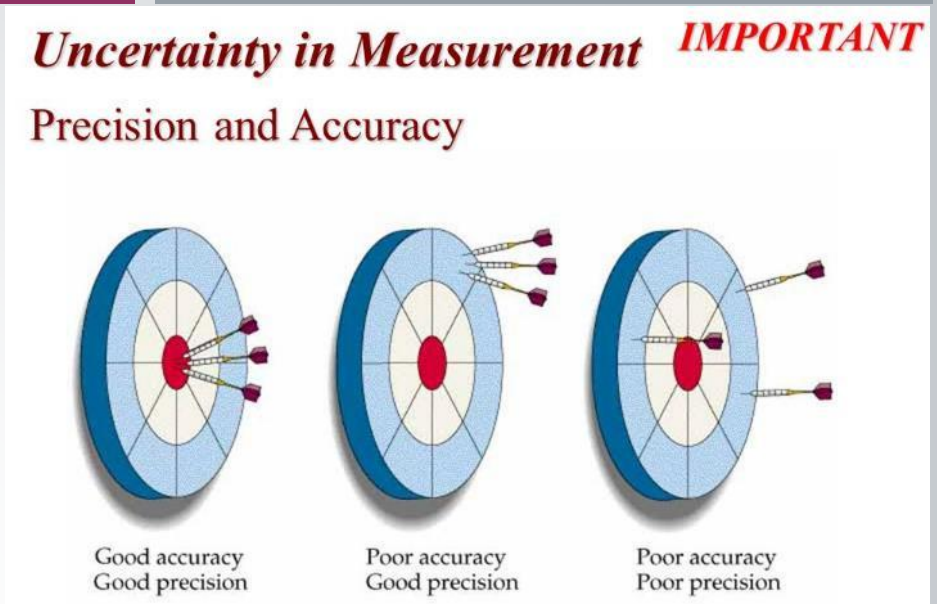
① Digits + ② Units



One number 4.72

sure Not sure

S.F.-minimum number of digits without loss of precision.



Accuracy and precision are best explained by darts throwing game

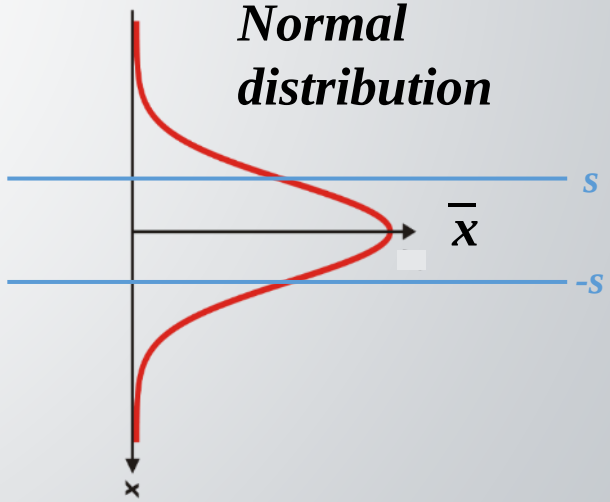
A group of number $x_1, x_2, \ldots x_{n-1}, x_n$

$\bar{x} \pm s, N = n$ $\quad s = \sqrt{\frac{\sum_i (x_i - \bar{x})^2}{N-1}}$ $\quad RSD = \frac{s}{\bar{x}} \times 100\%$

$\bar{x}$ -***Average***, **measure of accuracy**

s –***Standard deviation***, **measure of precision**

s_m –Standard deviation of mean



Chemical measurements

Propagation of error

S.F.

$$(\bar{x}_1 \pm s_1)\ \boxed{+\ /\ -\ /\ \times\ /\ \div}\ (\bar{x}_2 \pm s_2) = (\bar{x}_3 \pm s_3)$$

For the S.F. of $\bar{x}_3$

$\boxed{+\ /\ -}$ **Fewest decimal places**

$\boxed{\times\ /\ \div}$ **Smallest number of S.F.**

$\boxed{\log}$ Number of digits in **mantissa** of log x= number of significant figure (SF) in x:

log (x=5.403 × 10^{-8}) = -7.2674

4 SF 4 digits

For the S.F. of s_3

$\boxed{+\ /\ -}$ $s_3 = \sqrt{s_1^2 + s_2^2}$

$\boxed{\times\ /\ \div}$ $\%s_3 = \sqrt{(\%s_1)^2 + (\%s_2)^2}$

s Absolute error
$\%s$ Realtive error

Random Error

vs.

Systematic Error

Chemical measurements

① Digits + ② Units

② ppm 1 μg/mL; ppb 1 ng/mL

Making solution:

$$\underbrace{(M_{conc})(V_{conc})}_{\text{Moles taken from concentrated solution}} = \underbrace{(M_{dil})(V_{dil})}_{\text{Moles placed in dilute solution}}$$

Dilution formula

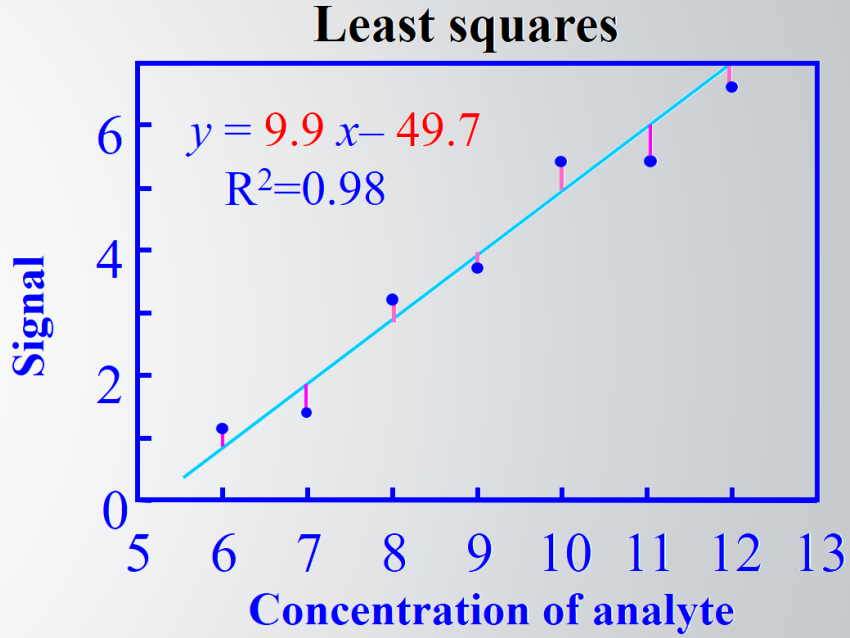
Limiting reagent: who consumed firstly

Chemical measurements

Linear regression: use the known to evaluate the unknown

1) External standard
2) Internal standard
3) Standard addition

Matrix Effect



3.1) Successive addition V ↑ x intercept = $[X]_i$

3.2) Multiple addition V is fixed, x intercept = $[X]_f$

$$[X]_i = \frac{V}{V_0}[X]_f$$

$$\frac{I_X}{I_{(x+s)}} = \frac{[X]_i}{[X]_f + [S]_f}$$

$$[X]_f = [X]_i \frac{V_o}{V}$$

$$[S]_f = [S]_i \frac{V_s}{V}$$

$$V = V_o + V_s$$

Chemical Equilibrium

1.Titration

- Curve
- Error
  - End point (Sudden change in measurement)
  - ↓ nearby
  - Equivalence point (Stoichiometric calculation)
- End point detection
  - ① Indicator
  - ② First derivative
  - ③ Gran plot

Chemical Equilibrium

1.Titration

- Curve
- Error
  - End point (Sudden change in measurement)
  - ↓ nearby
  - Equivalence point (Stoichiometric calculation)
- End point detection
- Types
  - ①Acid-base
  - ②Precipitation
  - ③Redox

Chemical Equilibrium

1.Titration

- ①Acid-base
  - Strong acid←NaOH
    $pH=7$
  - Weak acid(K_a)←NaOH (Buffer)
    pH=? at V_{NaOH}=0, ½V_e, V_e, >V_e
- ②Precipitation
  - Mohr T.
  - Fajans T.
  - Volhard T.

  $E = f(p[Ag^+])$ at V_e (K_{sp})

  $$E = E^o(Ag^+|Ag) - \frac{0.05916}{n} log \frac{1}{[Ag^+]}$$
- ③Redox

  E^o(Reductant pair); E^o(Oxidant pair)

  E=? at $V_{oxidant}$=½V_e, V_e, 2V_e

  $\Delta E = E^o(O.) - E^o(R.)$

Titration error: [Analyte]↑ e↓; ①K_a↑ e↓; ② K_{sp}↓ e↓; ③ΔE↑ e↓

Chemical Equilibrium

1.Titration

①Acid-base

Potentiometric

②Precipitation

$E = f\,(p[Ag^+])$ at V_e (K_{sp})

$$E = E^o\,(Ag^+|Ag) - \frac{0.05916}{n} log \frac{1}{[Ag^+]}$$

③Redox

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$\Delta E = E^o(O.) - E^o(R.)$

Chemical Equilibrium

2. Electro-analysis

Spontaneous -Galvanic cell

-Potentiometric (Titration)

Nernst Equation

$$E = E^{o}(Ag^{+}|Ag) - \frac{0.05916}{n} log \frac{1}{[Ag^{+}]}$$

Non-spontaneous -Elctroanalytical techniques

(Apply external potential)

- ❑ Voltammetry
- ❑ Electrogravimetric
- ❑ Coulometric

Chemical Equilibrium

2. Electro-analysis

Spontaneous -Potentiometric (Titration)

Precipitation

- ✓ Reagent conc.
- ✓ How to mix the solution
- ✓ Reaction temperature
- ✓ Digestion / aging time
- ✓ Washing liquid, cycle & removal
- ✓ Drying temperature

Non spontaneous-Elctroanalytical techniques

(A[illegible] external potential)

- ❑ Voltammetry
- ❑ Electrogravimetric
- ❑ Coulometric

Comparison

3. Gravimetric analysis

Final Exam for
125101 Analytical Chemistry 1

Term A: 14:00-17:00 June 24 (Tues.), 2025

Term B: 14:00-17:00 July 9 (Wed.), 2025

Final course grade = Final exam grade $\pm$ *e*

"***e***" includes the addition for good questions & deduction of homework missing; e $\leq$ 10 scores.

Final Exam for 125101 Analytical Chemistry 1

Instructions:

- The final exam will be without books or other material, except a calculator and dictionary. You may use a calculator and hardcopy dictionary of your own (after the checking by the invigilator).

(Do not use devices with large alphanumeric memory or of broadcasting capabilities.)

- Multiple Choice Questions (MCQ) with one correct answer for each question with 10~20 items.

MAY ALL OF YOU MAKE
SUCCESS IN THE FINAL EXAMS!